

SPACE X

SPATIAL INTELLIGENCE
SMART IMAGING

Manual



Version 1.0 - VST3 - macOS / Windows

What this is for

Space X is not a corrective tool. It is a way to give a sound its own layer in a mix that is already full.

Most stereo plugins are built to be safe. This one is built to be interesting. It moves things sideways, forwards, backwards and slightly out of phase - the area most engineers stay out of, either because nobody ever showed them how far it can be taken, or because they are afraid of what it does to mono. Used deliberately, that is exactly where the space is.

That comes out of the work. I have been mixing professionally since 2019 as a producer, mix and mastering engineer, on records and for television, among them The Voice. Alongside that I have been hired to rebuild and optimise recording and mixing templates for other engineers - Jeff Ellis and Mosty among them - so that their way of working would run on a different system. Sitting inside someone else's session is the fastest way to see where the decisions really are.

Two terms from Jeff Ellis' Mixer Brain course stuck, because they put a name on something I had been doing for years without one: the forbidden zone, and the fourth dimension. When a mix is dense and every frequency is taken, width and depth are the only room left. Space X is built to work in that room.

Think of it as an effect generator rather than a mastering tool. It takes something ordinary - a reverb return, a delay throw, a stack of backing vocals - and makes it worth listening to.

What it replaces

Everything below normally needs four or five separate plugins, and around 500 EUR:

Centre extraction (L / C / R)	a dedicated centre-separation plugin
Haas delay and micro pitch	a pitch / delay tool
Polarity, side gain, side width	a utility plugin - usually an ugly one
Mono check and correlation	a metering plugin
Distance and tone tilt	an EQ
Auto-pan and stereo movement	a panner

One window, one latency, one preset. That is most of the point.

Where to put it

It works best on anything that is already wet: backing vocals, ad-libs, a stack of harmonies, a reverb or a delay return. Throw tracks belong here too - those are usually single tracks rather than buses. A dry source is not off limits either: a lead vocal that is meant to be doubled works, and so do drums. The rule of thumb is simple - the more room effects sit in front of Space X, the further you can push it.

Position in the chain is a taste decision. Last is the usual place: shape whatever comes out of the chain. First also works when the incoming signal is already drenched in reverb or echo - then Space X shapes the effect itself rather than the source.

Throw tracks are the easiest win. They are 100 % wet by definition, nobody expects them to be mono-safe, and a small amount of Timewarp plus Vision turns a plain delay into a hook.

It is not a mastering tool. Do not put it on the mix bus and expect it to behave.

First result in one minute

1. Put it on a backing vocal bus or a throw track.
2. Click a Smart category at the top - Backings, Vocals, Pads, whatever fits.
3. Click the left dice. Listen. Click again. Keep clicking until something surprises you.
4. Too much? Pull Mix down to 70 %. Still too much? Turn one section off.
5. Check Mono at the bottom. If it collapses, lower Dimension Size a little.

That is the whole workflow. The knobs below exist for when you already know what you want.

The Smart engine

The two dice at the top randomise the plugin. The left one keeps every section on. The right one may switch sections off, which usually gives the cleaner result.

The six category chips change the rules behind the dice. Drums never gets a Haas delay and never a polarity flip. Pads gets slow movement and more RAYE. Backings is the widest of them. With a category selected, the left dice is greyed out - the category already decides which sections play.

The engine will not produce anything unusable. Loudness is compensated, extreme values are capped, and settings that would do nothing are skipped.

Tuned by ear. The dice are random; the rules they roll against are not. Every category, every wildcard and every exclusion was set by listening - which combinations carry a part and which ones quietly ruin it, where a polarity flip is a colour and where it is damage, how far Size can go before mono gives up. The settings that did not work were taken out before you ever rolled them. That is why you can hold the dice down and simply listen.

Lock (padlock)	this section is left alone by the dice and by Breathe
Breathe	switches modulation on in every section with fresh random depths
Undo / Redo	works on everything, including the dice
A / B	two versions side by side; the arrow copies one into the other

The signal path

Left to right, top to bottom. Polarity can sit at four different points in that chain - that is what the 1-4 buttons do.

IN

GALAXY

POLARITY

TIMEWARP

DIMENSION

HYPERDRIVE

VISION

RAYE

MIX

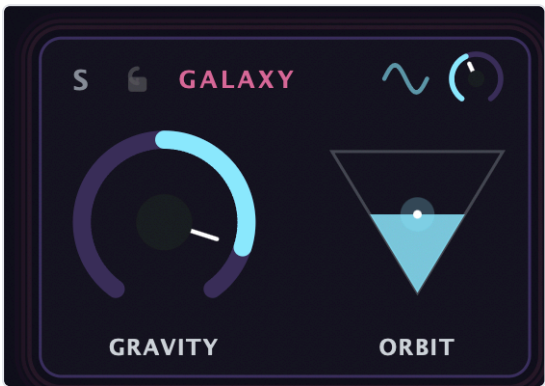
OUT

Latency is zero unless Galaxy is armed. Galaxy needs a analysis window to separate the centre, so switching it on costs a little delay. Everything else runs in real time.

On every section

S	solo - hear this section on its own
Padlock	the dice leave this section alone
Section name	click it to switch the section on or off
Wave icon	modulation on / off for this section
Small knob	modulation depth

GALAXY

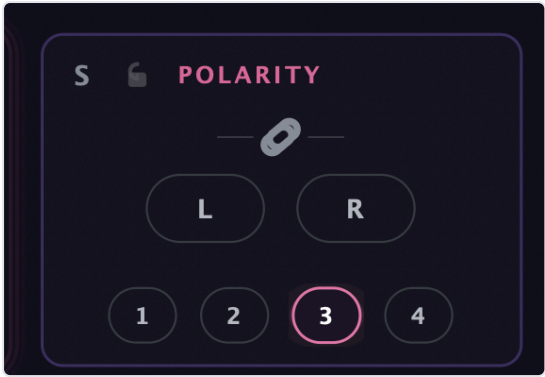


Pulls the centre out of the stereo image and treats centre and sides separately. This is the part that costs latency, and the part that makes everything after it sound different.

Gravity	how hard the centre is separated from the sides
Orbit	down keeps only the centre, up keeps only the sides, middle is the original

On a backing stack, pull Orbit up. The lead stays where it is and the backings open up around it without getting louder.

POLARITY



Flips the phase of one channel. The most drastic thing in the plugin and the one that makes the biggest difference.

L / R	flip the left or the right channel
Link	both at once
1 - 4	where in the chain the flip happens

A flip almost always breaks mono compatibility. On a throw or an effect return that is fine - nobody sums those to mono and expects them to survive. On anything that carries the song, check Mono first.

The four positions matter more than they look. Flipping before Galaxy changes what Galaxy considers the centre. Flipping after Vision only affects the final image. Try 1 and 3 on the same setting - they are two different sounds.

TIMEWARP



Tiny differences between left and right in time and in pitch. This is the classic width trick, and the reason two plugins used to be needed.

Drift	delays one side by a few milliseconds (Haas). Left or right
Shift	micro pitch offset between the channels, in cents
Balance	evens out the loudness shift Drift causes

DIMENSION

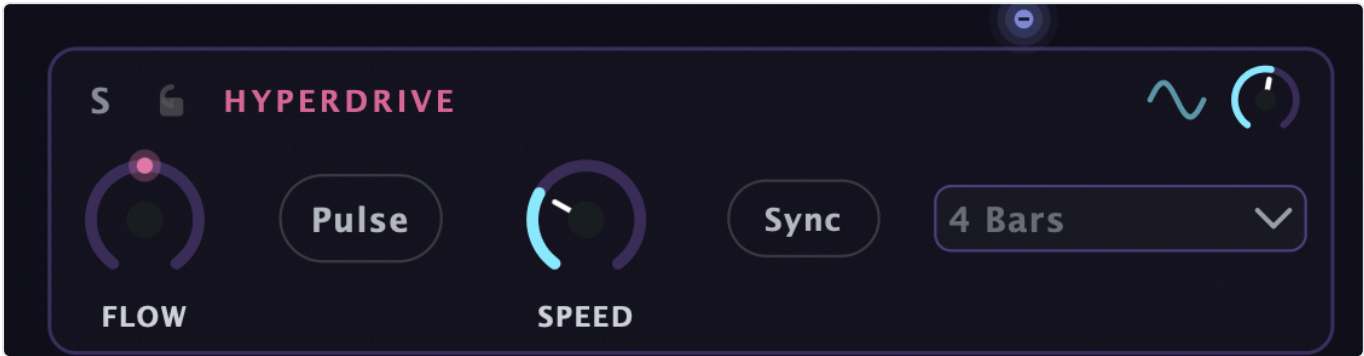


Classic mid / side width. The section you reach for when the result is nearly right and just needs to be wider or narrower.

Size	stereo width. Below 100 % narrower, above 100 % wider
Boost	adds level to the sides

Size above roughly 150 % together with Drift is where mono starts to suffer. The plugin already limits the worst combinations, but your ears decide.

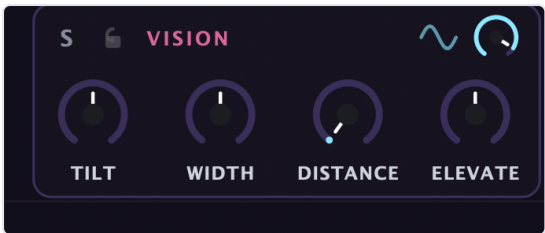
HYPERDRIVE



Slow automatic movement through the stereo field. Not a tremolo - the level stays put, only the position moves.

Flow	how far the sound travels left and right
Pulse	a smoother, pulse-like movement instead of a sine
Speed	movement rate in Hz
Sync	lock the rate to the host tempo
Bars	the note length used while Sync is on

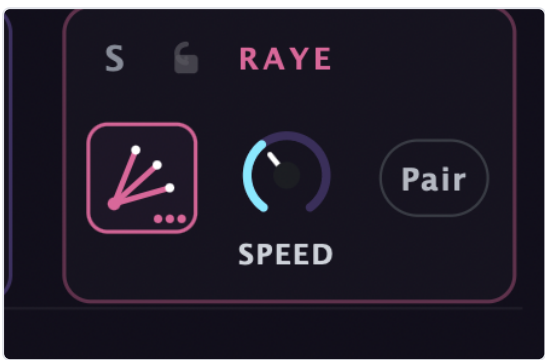
VISION



Where the finished picture sits and how far away it is. This is the section that replaces an EQ move you would otherwise do by hand.

Tilt	shifts the whole image left or right
Width	final width of the image
Distance	further away means a softer top end and a little less level
Elevate	gentle tone tilt. Up is brighter, down is darker

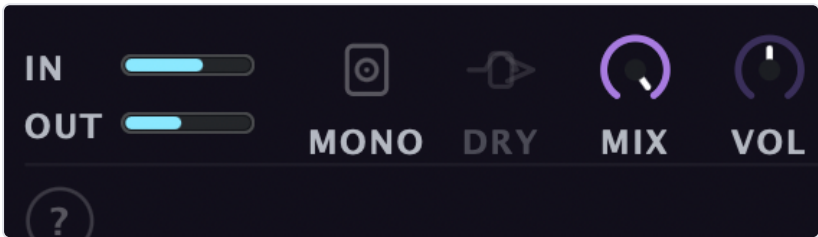
RAYE



A soft phaser that moves the stereo image rather than the tone. Subtle on purpose - it is movement, not an effect you should be able to name.

Strength	click to step through Light, Medium, Strong and Off
Speed	how fast the movement cycles
Pair	follow Hyperdrive at half its speed

Output



IN / OUT	input and output level
Mono	listen to the result in mono
Dry	while in mono, compare with the unprocessed input
Mix	blend the processed sound with the original
Vol	output trim, plus or minus 6 dB
Correlation	right of centre is mono-safe, left of centre cancels in mono

Frequency band



Limits the width processing to a frequency range. Outside the band nothing is widened. Drag an edge to resize, drag the middle to move, drag up and down or scroll to widen.

Power	band on / off
Band	the range in which Dimension and Vision are allowed to work

Set the lower edge to around 200 Hz and the low end stays mono and solid while everything above it opens up. This is the single most useful setting in the plugin.

The starfield



The picture on the left is not decoration only. Every object reacts to what the plugin is actually doing - the planet rises with Gravity, the star lines lean with Tilt, the golden lines are RAYE.

Click the field to switch the scope on or off. Click the sun or the moon to switch between the full view and the clean meter view. The gear opens the display settings.

Goniometer	the classic vectorscope - a vertical line is mono, a circle is wide
Scene	Space shows everything, Stars is the quiet meter view
Speed / Density	how fast and how busy the field is
Stars	switch the twinkling stars off

Themes and layout

The menu holds seven themes and three layouts. Layout changes how much framing the sections get: 3D has boxes and fill, Flat has neither, Outline has a thin frame and no fill. None of it changes the sound.

Five settings that work

Backing stack that will not fit

Category Backings, right dice. Then Orbit up, Dimension Size around 120 %, frequency band from 200 Hz. The lead keeps the centre, the backings take everything around it.

Delay throw with an identity

Timewarp Drift at 8-12 ms, Shift around 6 cents, RAYE on Light. The throw stops sounding like a delay and starts sounding like a decision.

Reverb return with depth

Vision Distance up, Elevate down a little, Dimension Size around 130 %. The reverb moves behind the dry signal instead of on top of it.

Ad-libs that appear and disappear

Hyperdrive Flow around 40 %, Sync on, 4 bars. Slow enough that nobody notices the movement, fast enough that the part never sits still.

The forbidden one

Polarity L on, position 1, Mix around 60 %. Mono will not survive it. On a throw in a dense chorus it is the widest thing you own.

Clicks worth knowing

Double-click a knob	back to default
Cmd / Ctrl + drag	fine adjustment
Click the section name	section on / off
Click the logo	bypass
? bottom left	shows what the mouse is over

Credits

Space X started as a way to stop opening five plugins for one idea. Nothing in it happened by accident: the order of the chain, the four polarity positions, the limits behind the dice - every one of those is a decision that was made by listening, and then made again.

Thanks to Jeff Ellis, whose words for the forbidden zone and the fourth dimension gave the idea a name. If Space X earns its place on your bus, a coffee is always welcome.

[Paul Misty](#) · [Mistycatstudios](#) · info@paulmisty.com · paulmisty.com · [Germany](#)